

day 19 | 251010 F

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def bisection(function, lower_guess, upper_guess,
tolerance=2**(-32)):
    midpoint = (lower_guess + upper_guess)/2
    while upper_guess - lower_guess > tolerance:
        if function(lower_guess)*function(midpoint)<0:
            upper_guess = midpoint
            midpoint = (lower_guess + upper_guess)/2
        elif function(midpoint)*function(upper_guess)<0:
            lower_guess = midpoint
            midpoint = (lower_guess + upper_guess)/2
        elif function(lower_guess)*function(midpoint)>0 and
function(midpoint)*function(upper_guess)>0:
            print('no unique root in that bracket')
            break
    return(midpoint)

def newton(f, df, guess, tolerance = 2**(-32)):
    x = guess
    n = 0
    while abs(f(x)) > tolerance:
        x = x - f(x)/df(x)
        n += 1
    return(x, n)

def secant(f, guess, delta, tolerance = 2**(-32)):
    x0 = guess
    x1 = x0 + delta
    n = 0
    while abs(f(x1))>tolerance:
        x1 = x1 - (x1-x0)/(f(x1)-f(x0))*f(x1)
        n += 1
    return(x1, n)
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In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
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In [10]: def bisection(function, lower_guess, upper_guess, tolerance=2**(-32)):
midpoint = (lower_guess + upper_guess)/2
n = 0
while upper_guess - lower_guess > tolerance:
    if function(lower_guess)*function(midpoint)<0:
        upper_guess = midpoint
        midpoint = (lower_guess + upper_guess)/2
    elif function(midpoint)*function(upper_guess)<0:
        lower_guess = midpoint
        midpoint = (lower_guess + upper_guess)/2
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        elif function(lower_guess)*function(midpoint)>0 and function(midpoint)*function(upper_guess)<0:
            print('no unique root in that bracket')
            break
        n = n+1
    return(midpoint, n)

def newton(f, df, guess, tolerance = 2**-32):
    x = guess
    n = 0
    while abs(f(x)) > tolerance:
        x = x - f(x)/df(x)
        n += 1
    return(x, n)

def secant(f, guess, delta, tolerance = 2**-32):
    x0 = guess
    x1 = x0 + delta
    n = 0
    while abs(f(x1))>tolerance:
        x1 = x1 - (x1-x0)/(f(x1)-f(x0))*f(x1)
        n += 1
    return(x1, n)

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In [11]: def f(x):
        return(5*np.exp(-x)+x-5)

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In [20]: fig, ax = plt.subplots()

        x = np.linspace(-1,10)
        y = f(x)

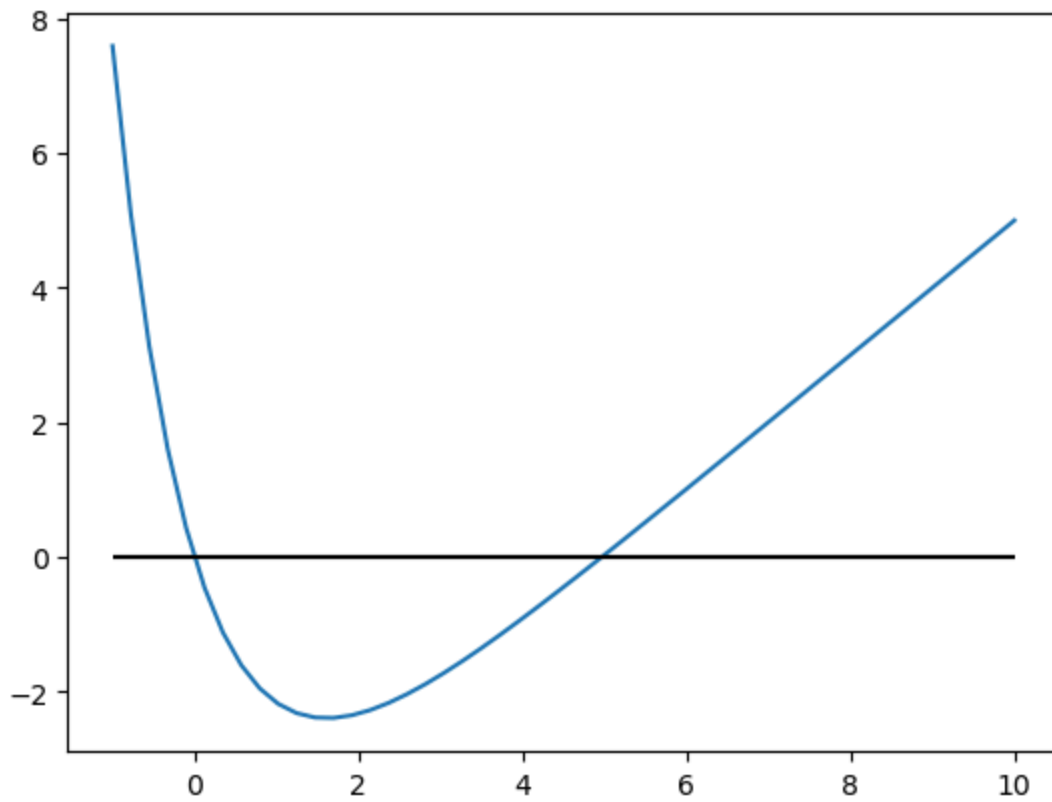
        ax.plot(x,y)
        ax.hlines(0, x[0], x[-1], 'k')

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Out[20]: <matplotlib.collections.LineCollection at 0x760efd9d8f50>

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In [13]: bisection(f, 4, 6)
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Out[13]: (4.9651142318034545, 33)
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In [14]: def df(x):  
         return(-5*np.exp(-x)+1)
```

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In [18]: newton(f, df, 0)
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Out[18]: (0, 0)
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In [21]: secant(f, 10, 1)
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Out[21]: (4.965114231761079, 7)
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In [ ]:
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